

Exam 1

● Graded

Student

Cayden Gosseck

Total Points

92.75 / 100 pts

Question 1

Vocabulary/Theory

3 / 3 pts

✓ + 3 pts Correct

Question 2

Vocabulary/Theory

5 / 5 pts

✓ - 0 pts Correct

Question 3

Vocabulary/Theory

3 / 3 pts

✓ + 3 pts Correct

Question 4

Vocabulary/Theory

3 / 3 pts

✓ + 3 pts Correct

Question 5

Vocabulary/Theory

3 / 3 pts

✓ - 0 pts Correct

Question 6

Vocabulary/Theory

3 / 3 pts

✓ + 3 pts Correct

Question 7

Vocabulary/Theory

0 / 3 pts

✓ + 0 pts Incorrect

Question 8

Reading UML/Java Code

17 / 17 pts

8.1 Reading UML/Java Code

4 / 4 pts

✓ - 0 pts Correct

8.2 Reading UML/Java Code

4 / 4 pts

✓ - 0 pts Correct

8.3 Reading UML/Java Code

5 / 5 pts

✓ - 0 pts Correct

8.4 Reading UML/Java Code

4 / 4 pts

✓ - 0 pts Correct

Question 9

Vocabulary/Theory

5 / 5 pts

✓ - 0 pts Correct

Question 10

Java Code

19.75 / 20 pts

10.1 Java Code

11.75 / 12 pts

Convenience Constructor

✓ - 0.25 pts Constructor should update number of books in a static manner

10.2 Java Code

8 / 8 pts

✓ - 0 pts Correct

Question 11

Vocabulary/Theory

3 / 3 pts

✓ + 3 pts Correct

Question 12

Vocabulary/Theory

3 / 3 pts

✓ + 3 pts Correct

Question 13

Vocabulary/Theory

3 / 3 pts

✓ + 3 pts Correct

Question 14

Vocabulary/Theory

5 / 5 pts

✓ - 0 pts Correct

Question 15

Vocabulary/Theory

2 / 3 pts

✓ - 1 pt Missing one correct item

Question 16

Vocabulary/Theory

0 / 3 pts

✓ + 0 pts Incorrect

Question 17

Vocabulary/Theory

3 / 3 pts

✓ + 3 pts Correct

Question 18

Vocabulary/Theory

3 / 3 pts

✓ + 3 pts Correct

Question 19

Vocabulary/Theory

3 / 3 pts

✓ + 3 pts Correct

Question 20

Vocabulary/Theory

3 / 3 pts

✓ + 3 pts Correct

Question 21

Vocabulary/Theory

3 / 3 pts

✓ + 3 pts Correct

Q1 Vocabulary/Theory

3 Points

How are static members different from instance members?

- ☐ An instance member can be accessed outside of the class, but a static member cannot
- ☐ They are not different in any way
- ☒ Accessing static members is different than instance members because they belong to the class, not an instance
- ☐ Accessing static members is the exact same as instance members, even though they are shared

Q2 Vocabulary/Theory

5 Points

Provide the recommended visibility modifier that Java convention dictates is usually appropriate for a class's methods.

public

Q3 Vocabulary/Theory

3 Points

In Java, the set of ____ define the behavior of an entity.

- ☒ methods
- ☐ data members
- ☐ properties
- ☐ values

Q4 Vocabulary/Theory

3 Points

To prevent direct modification of a data member, the member should have a visibility of ____.

- ☒ private
- ☐ default
- ☐ public
- ☐ final

Q5 Vocabulary/Theory

3 Points

Modification of encapsulated data members require ____ and ____ methods.
Choose all correct answers.

☐ fetch/update

☒ accessor/mutator

☐ retrieve/change

☒ get/set

☐ instance/static

Q6 Vocabulary/Theory

3 Points

OOP entities and their properties, including any relationships to other entities, can be visually represented using ____ that uses symbols like rectangles, lines/arrows, and other shapes.

- ☐ Uniform Modeling Language
- ☒ Unified Modeling Language
- ☐ Unified Model Language
- ☐ Uniform Model Language

Q7 Vocabulary/Theory

3 Points

How does Java know what constructor to use when invoked? Choose the most correct answer.

- ☐ Using method matching
- ☒ Order of parameters only
- ☐ Number of parameters only
- ☐ JVM chooses any implemented constructor

Q8 Reading UML/Java Code

17 Points

Answer the following questions using this class UML:

Scooter
<pre>-manufacturer: String -model: String -displacement: double -seat: Seat -maxSpeed: double</pre>
<pre>+Scooter() +Scooter(man: String, mod: String) +getManufacturer(): String +setManufacturer(man: String): void +getModel(): String +setModel(mod: String): void +getDisplacement(): double +setDisplacement(dis: double): void +getSeat(): Seat +setSeat(seat: Seat): void +getMaxSpeed(): double</pre>

Q8.1 Reading UML/Java Code

4 Points

Assume you are working in a `main` method of a class that has access to the `Scooter` class.

Create a default instance of a `Scooter` named `scoot1`.

```
Scooter scoot1 = new Scooter();
```

Q8.2 Reading UML/Java Code

4 Points

Assume you are working in a `main` method of a class that has access to the `Scooter` class.

Set the manufacturer of `scoot1` (from the problem asking you to create a default `Scooter` instance) to "Yamaha" and the model to "XMAX".

Print the `scoot1`'s model without any descriptive text.

```
scoot1.setManufacturer("Yamaha");  
scoot1.setModel("XMAX");  
System.out.print(scoot1.getModel());
```

Q8.3 Reading UML/Java Code

5 Points

Assume you are working in a `main` method of a class that has access to the `Scooter` class.

Create an instance of `Scooter` named `scoot2` with a manufacturer of "Wolf", the model to "RX-50", and displacement to 49.5 cc.

This should be done in the least number of steps possible.

```
Scooter scoot2 = new Scooter("Wolf", "RX-50");  
scoot2.setDisplacement(49.5);
```

Q8.4 Reading UML/Java Code

4 Points

Assume you are working in a `main` method of a class that has access to the `Scooter` class.

Print out the maximum speed of a `Scooter`. There should be no descriptive text.

```
System.out.println(Scooter.getMaxSpeed());
```


Q9 Vocabulary/Theory

5 Points

Provide the recommended visibility modifier that Java convention dictates is usually appropriate for a class's constructors.

public

Q10 Java Code

20 Points

Answer the following questions:

Q10.1 Java Code
12 Points

Book
-title: String -author: String -year: int <u>-totalNumBooks: int</u>
+Book() +Book(title: String, author: String, year: int) +getSynopsis(): String <u>+getTotalNumBooks(): int</u>

Create a `Book` class to represent individual books that could be housed in a library. `Book`s have a title, author, and year published as well as a method to retrieve a synopsis (short description) of the book.

An incomplete (excludes some items like accessors/mutators, etc.) UML diagram for `Book` is provided.

Implement **ONLY** the following in the class:

- The class header
- Declare all data members with appropriate visibility, modifiers, and data type.
- A constructor that takes a title, author, and year as parameters and appropriately uses those values. This constructor should update any other relevant data members.
- Accessor and mutator methods for the title data member. You may omit implementations for all other data members, **but assume they exist**.
- A `getSynopsis()` method that returns: "title by author (year)" with the appropriate values for the specific book instance (include the parentheses around the year).

For all items, use best practices according to Java/OOP convention.

```
public class Book {  
    private String title;  
    private String author;  
    private int year;  
    private static int totalNumBooks;  
  
    public Book(String title, String author, int year) {  
        setTitle(title);  
        setAuthor(author);  
        setYear(year);  
        totalNumBooks++;  
    }  
}
```

```
public String getTitle() {  
    return title;  
}  
  
public void setTitle(String title) {  
    this.title = title;  
}  
  
public String getSynopsis() {  
    return String.format("%s by %s(%d)", getTitle(), getAuthor(),  
getYear());  
}  
}
```

Q10.2 Java Code

8 Points

Book
<pre>-title: String -author: String -year: int -totalNumBooks: int</pre>
<pre>+Book() +Book(title: String, author: String, year: int) +getSynopsis(): String +getTotalNumBooks(): int</pre>

Write the main method for a test program that creates an array of `Book`s to represent a personal library.

One of the books should have a title of *First Impressions* by Jane Austen, which was originally published in 1796. Another book should have a title of *Animal Farm* by George Orwell which was first published in 1945. The final book should have a title of *Catch 22* by Joseph Heller which was first published in 1961.

First Impressions was renamed *Pride and Prejudice* and re-published in 1813. Update the instance associated with that specific `Book` to reflect those changes **AFTER** its initial creation.

Print the synopsis for each book in the array using iteration. Additionally, print the total number of `Book`s currently in the library using the `Book` class.

Assume best practices are being followed for all classes.

```
public static void main(String[] args) {

    Book bookOne = new Book("First Impressions", "Jane Austen", 1796);
    Book bookTwo = new Book("Animal Farm", "George Orwell", 1945);
    Book bookThree = new Book("Catch 22", "Joseph Heller", 1961);

    bookOne.setTitle("Pride and prejudice");
    bookOne.setYear(1813);

    Book[] bookArray = {bookOne, bookTwo, bookThree};

    for(int i=0; i < bookArray.length; i++) {
        System.out.println(bookArray[i].getSynopsis());
    }
}
```

```
        System.out.printf("\nTotal Num. Books: %d",  
Book.getTotalNumBooks());  
    }
```

Q11 Vocabulary/Theory

3 Points

The ____ relationship is represented by the filled diamond shape in UML.

- ☐ aggregation
- ☐ association
- ☐ binary
- ☒ composition

Q12 Vocabulary/Theory

3 Points

In Java, those entities are modeled in a(n) ____, a template (or blueprint) that defines the state and behavior.

- ☐ item
- ☐ object
- ☒ class
- ☐ thing

Q13 Vocabulary/Theory

3 Points

The ____ relationship is represented by the hollow diamond shape in UML.

- ☒ aggregation
- ☐ binary
- ☐ association
- ☐ composition

Q14 Vocabulary/Theory

5 Points

Provide the recommended visibility modifier that Java convention dictates is usually appropriate for a class's data members.

private

Q15 Vocabulary/Theory

3 Points

Which class components does the `this` keyword provides access to when used in a Java class? Choose all correct answers.

☐ methods in other classes

☒ data members

☒ methods in the class

☐ constructors

Q16 Vocabulary/Theory

3 Points

In Java, the set of data members and their current values model the ____ of an entity.

☒ properties

☐ behavior

☐ existence

☐ state

Q17 Vocabulary/Theory

3 Points

Aggregation models the "___" relationship between classes.

- ☐ is a
- ☐ is not like a
- ☐ is kind of
- ☒ has a

Q18 Vocabulary/Theory

3 Points

What is the difference between the aggregation and composition relationships?

- ☐ Aggregation requires that the aggregated object's existence depends on the aggregating object
- ☐ Aggregation models the "has a" relationship and composition models an "is a" relationship
- ☐ There is no difference between them
- ☒ Composition requires that the aggregated object's existence depends on the aggregating object

Q19 Vocabulary/Theory

3 Points

A(n) ___ represents a real-world entity that can be distinctly identified.

- ☒ object
- ☐ thing
- ☐ class
- ☐ item

Q20 Vocabulary/Theory

3 Points

How are constructors invoked outside of the class in Java? Choose the most correct answer.

- ☒ `new` keyword
- ☐ `public` keyword
- ☐ `this` keyword
- ☐ Developer writing a test program

Q21 Vocabulary/Theory

3 Points

What is the purpose of a static member?

- ☐ To store a value specific to an instance of the class
- ☐ Only to allow main methods to know how many of something exists
- ☒ To store a value that is shared among all instances of a class
- ☐ To represent a value that can never change in a class